Spell Check

(part of IR pre-processing steps)

- 2 key problems -
  - → how do we map variants of words to their root forms
  - → how to overcome spelling errors

- why correcting spelling errors is impt.
  - → human errors in typing (search engines)
  - → OCR
  - → speech-to-text

- why are spelling errors made?
  - → homophones
  - → no neat mapping btw. structure & pronunciaⁿ of words.

- spelling error categories

Non-words	Words
- seperate for separate	- dessert for desert
	- piece for peace

Typographic errors (homologous errors due to keyboard)

cognitive errors

- how to recover? → edit distance, keyboard, pronunciation, context, syntax, source-specific issues (like OCR)

- 2 classes of sys. → spelling error detection
  ↓
  spelling error correction

- typing errors → considering single error mis-spellings
  → insertion, deletion, substitution, transposition

- OCR errors → substitution, multi-substitution (framing errors), space deletion, space insertion, failures

- spell-check for non-words
  → detection / correction
  → vector-space approach
  → bigrams
  → syntactic pattern recognition
  → evaluation
  → diff. dimensions

- co-occurrence → two or more words appear together within a certain context

- collocaⁿ → words that frequently & naturally occur together.

- bayesian model

$$\underbrace{p(c|t)}_{\text{posterior}} = \frac{\overbrace{p(t|c)}^{\text{likelihood}} \cdot \overbrace{p(c)}^{\text{prior}}}{\underbrace{p(t)}_{\text{evidence}}}$$

t → typo

c → candidat